INCH-POUND
MIL-DTL-16377/80A
w/AMENDMENT 1
25 July 2022
SUPERSEDING
MIL-DTL-16377/80A
30 June 2014

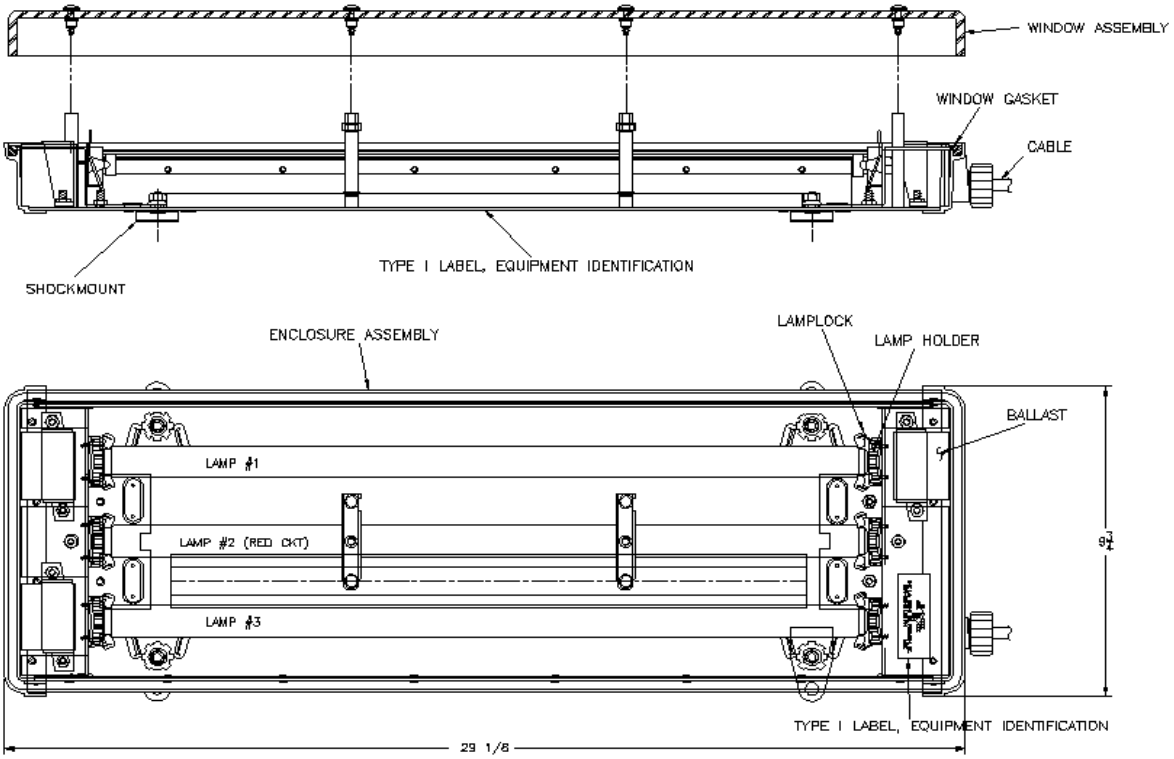
DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; T8 FLUORESCENT, GENERAL LIGHTING, 51 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 333.1T8, 333.2T8, 341.1T8, 341.2T8, 333.1T8RC, 341.4T8, 342.2T8, AND 342.3T8

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377. If specified by the acquisition activity, solid state lighting (SSL) fixtures designed and manufactured in accordance with MIL-DTL-16377/12 may be substituted for fixtures manufactured in accordance with this specification sheet.

All dimensions are in inches. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.



NOTES:

1. Symbols 333.2T8 and 342.3T8 use identical construction with flush mounting flange not shown.
2. Symbols 341.1T8 and 341.2T8 use identical construction with lens mounted black louvers not shown.
3. Symbols 333.1T8RC and 341.4T8 use identical construction with center lamp shield not shown.

FIGURE 1. Typical view of symbols 333.1T8 and 342.2T8.

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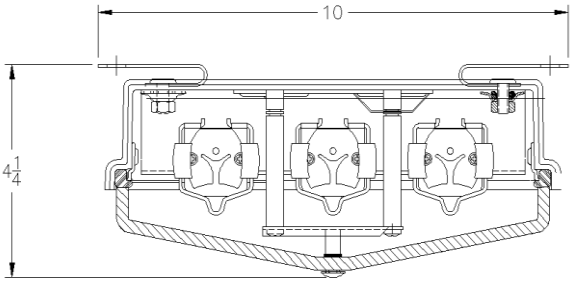
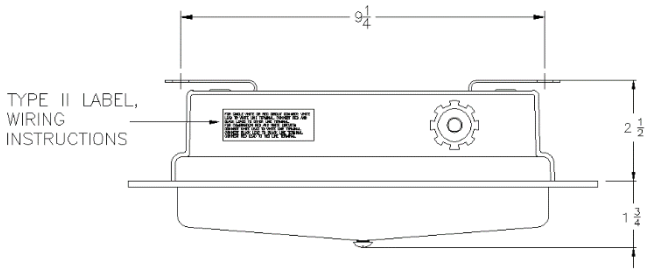


FIGURE 2. Typical section for symbols 333.1T8 and 342.2T8.



NOTE:

1. Type II label required for all fixtures.

FIGURE 3. Typical section for symbols 333.2T8 and 342.3T8.

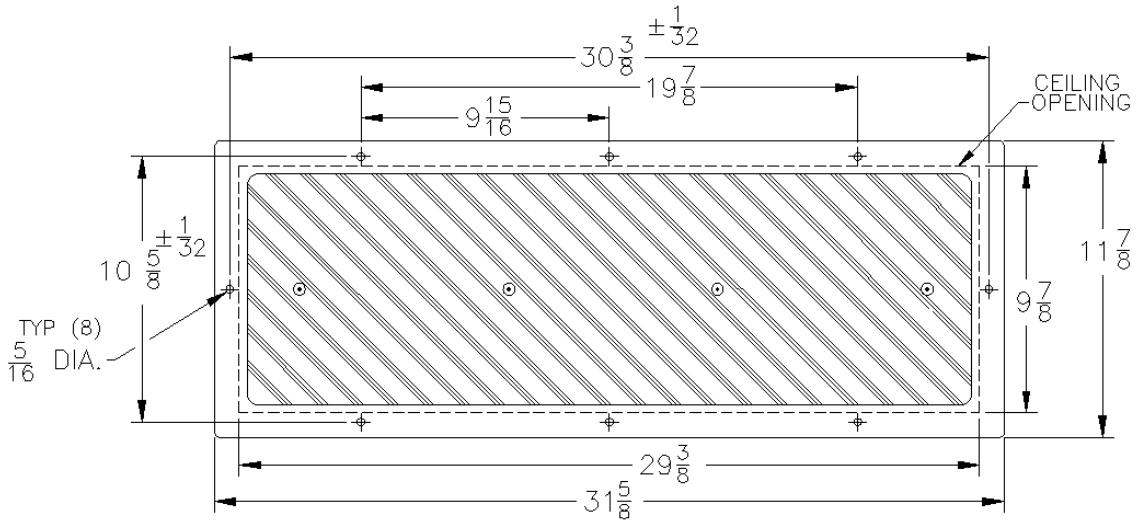


FIGURE 4. Flush mounting dimensions for symbols 333.2T8, 341.4T8, and 342.3T8.

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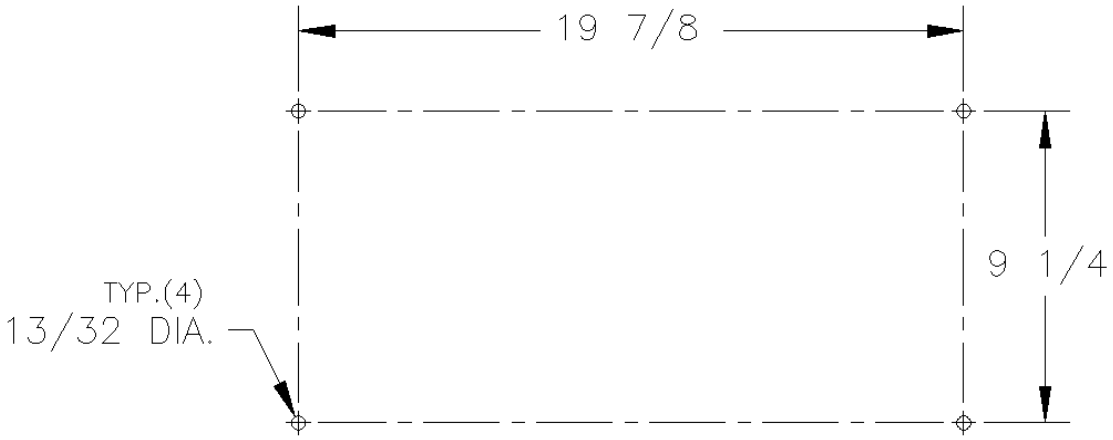
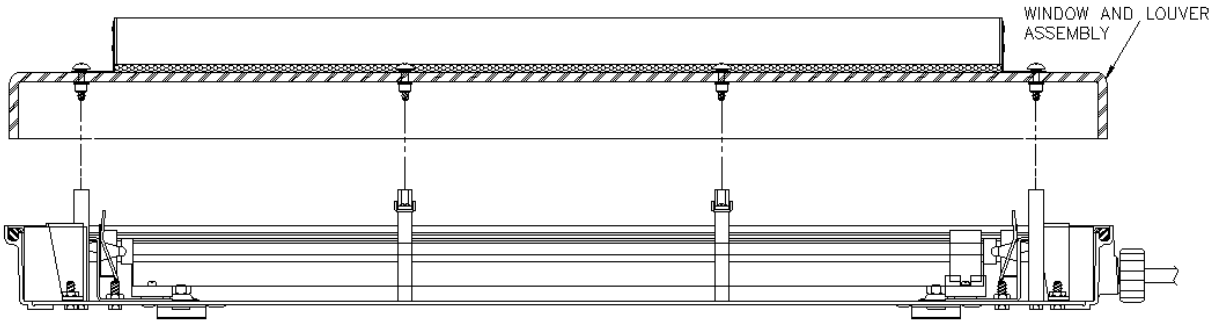


FIGURE 5. Standard mounting dimensions utilizing shock mounts for symbols 333.1T8, 341.1T8, 333.1T8RC, and 342.2T8.



NOTE:

1. Symbol 341.2T8 has identical construction with flush mounting flange not shown.

FIGURE 6. Typical view of symbol 341.1T8 with lens mounted black louvers.

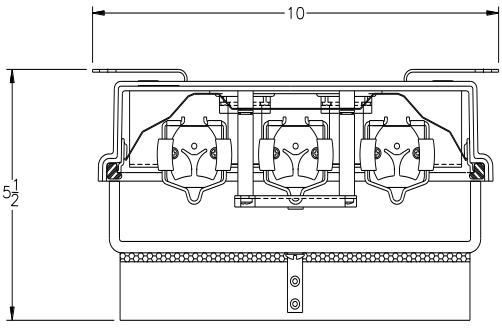
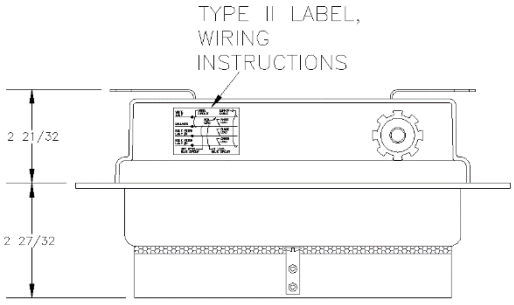


FIGURE 7. Typical section for symbol 341.1T8 with lens mounted black louvers.



NOTE:

1. Type II label required for all fixtures.

FIGURE 8. Typical section for symbol 341.2T8 with lens mounted black louvers.

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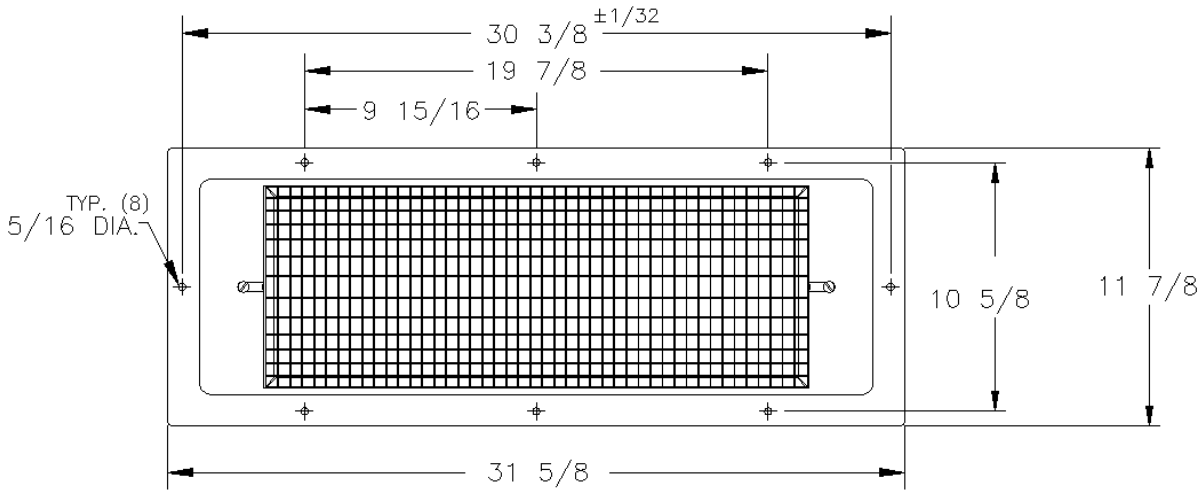
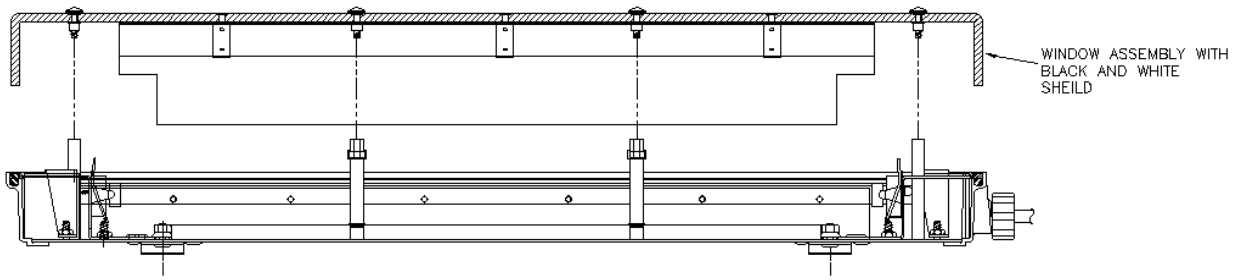


FIGURE 9. Flush mounting dimensions for symbol 341.2T8 with lens mounted black louvers.

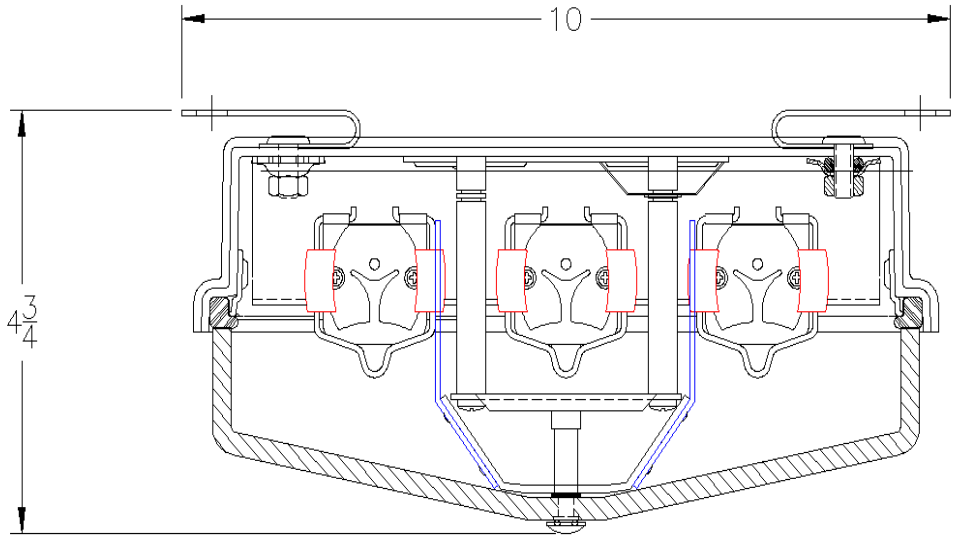


NOTE:

1. Symbol 341.4T8 has identical construction with flush mounting flange not shown.

FIGURE 10. Typical view of symbol 333.1T8RC with center lamp shielded.

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NOTE:

1. Symbol 341.4T8 has identical construction with flush mounting flange not shown.

FIGURE 11. Typical section for symbol 333.1T8RC with center lamp shielded.

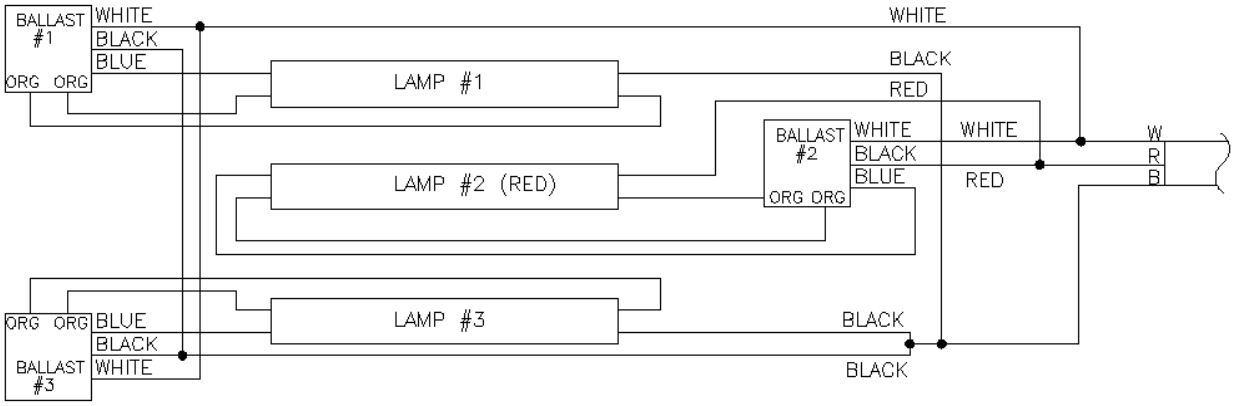


FIGURE 12. Typical wiring diagram.

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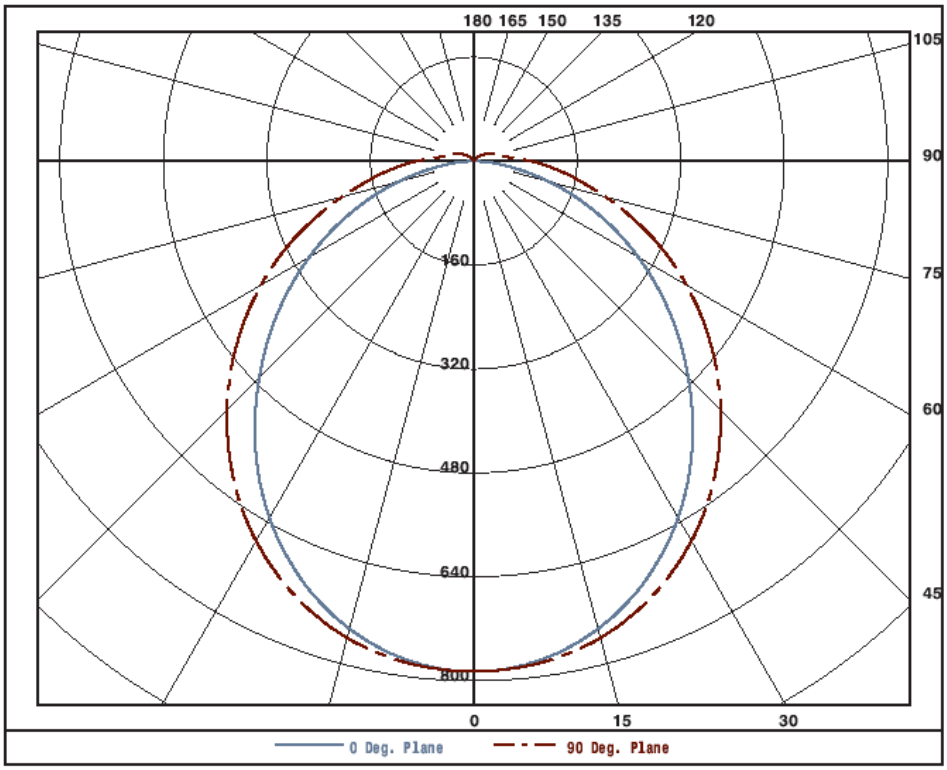


FIGURE 13. Candela distribution curves, white diffusing window, symbols 333.1T8 and 333.2T8.

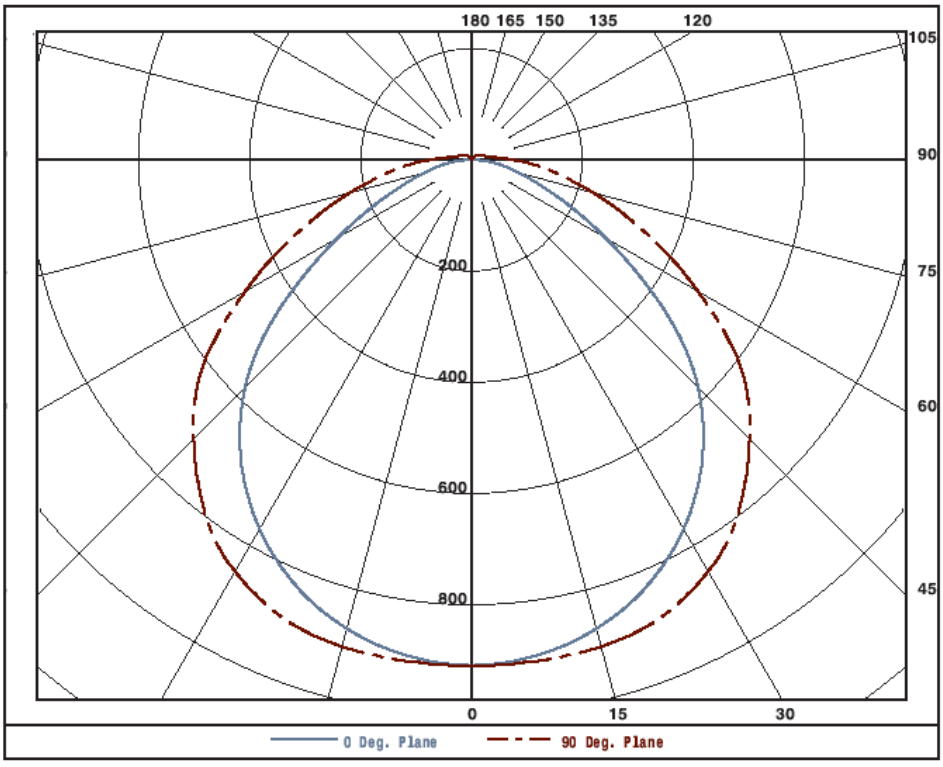


FIGURE 14. Candela distribution curves, prismatic window, symbols 342.2T8 and 342.3T8.

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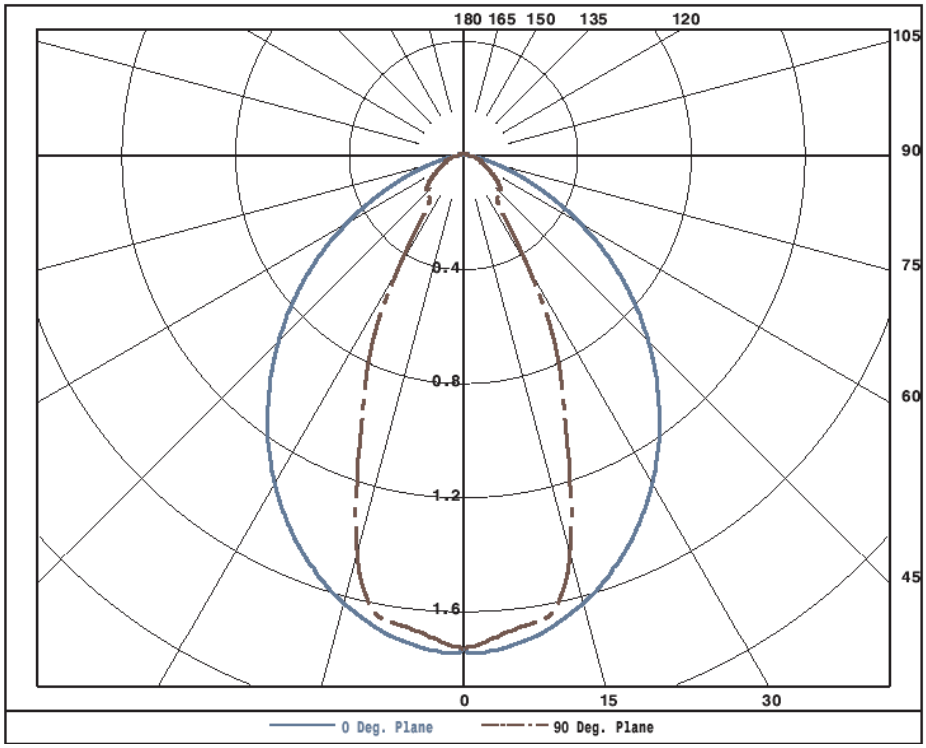


FIGURE 15. Candela distribution curves (red mode), one red filter center lamp energized, two white lamps off, symbols 333.1T8RC and 341.4T8.

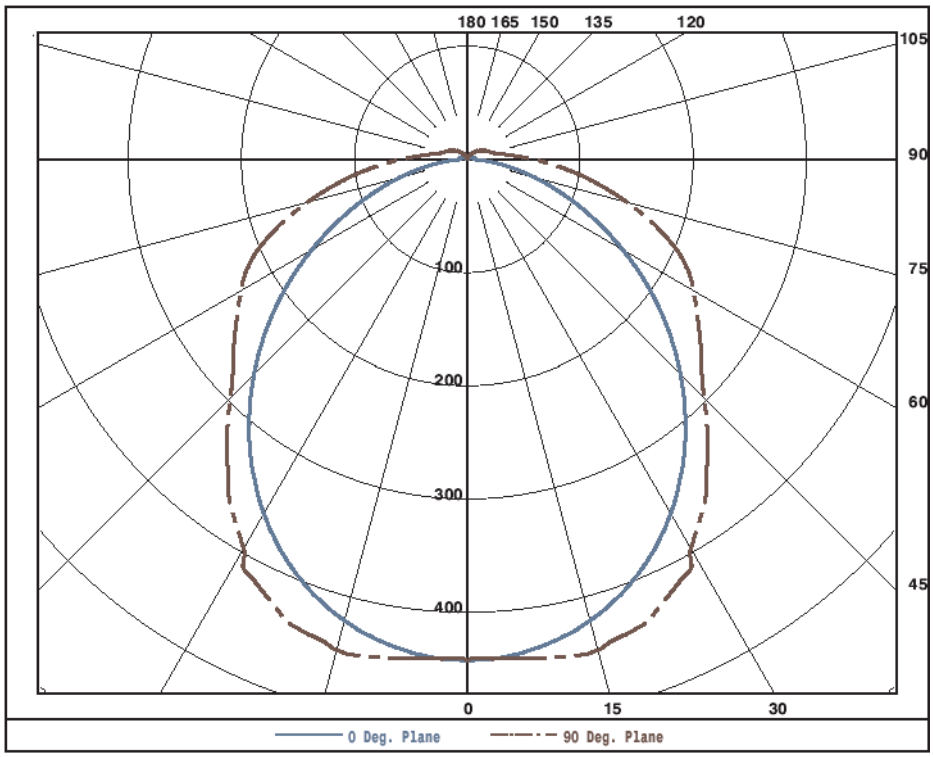


FIGURE 16. Candela distribution curves (white mode), two white lamps energized, red center lamp off, symbols 333.1T8RC and 341.4T8.

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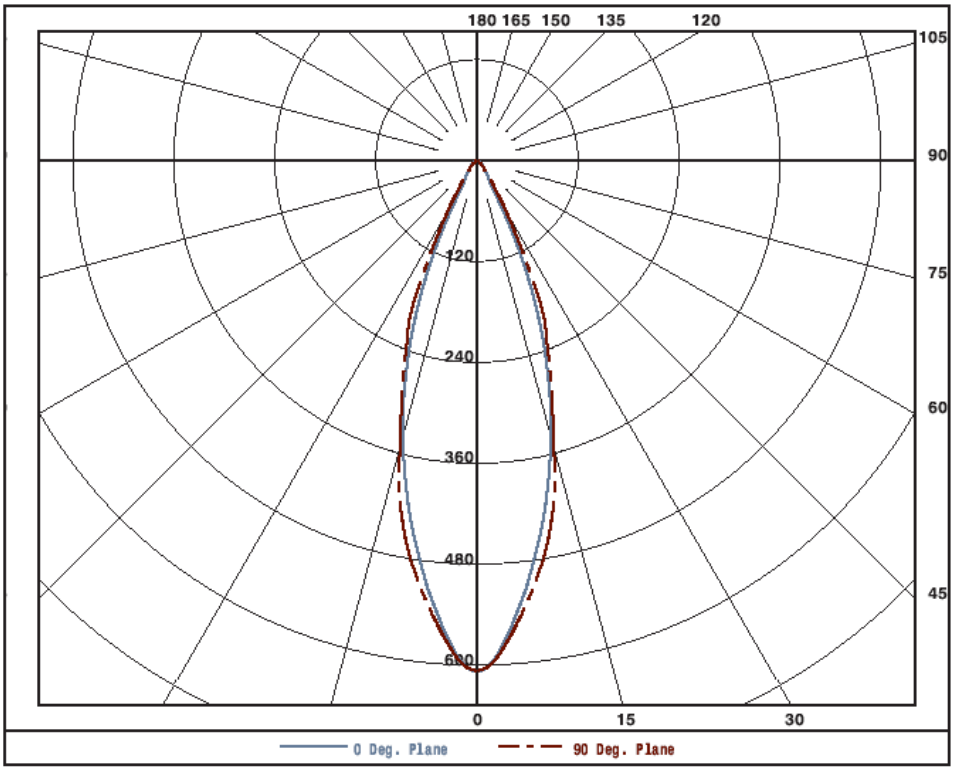


FIGURE 17. Candela distribution curves, three white lamps, symbols 341.1T8 and 341.2T8.

TABLE I. Numerical candela distribution, white diffusing window, symbols 333.1T8 and 333.2T8.

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
0	786	786
5	781	783
15	745	760
25	677	709
35	582	634
45	469	540
55	353	436
65	234	323
75	120	211
85	27	124
90	9	89

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
95	8	62
105	6	39
115	4	26
125	3	17
135	0	10
145	0	3
155	0	0
165	0	0
175	0	0
180	0	0

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TABLE II. Numerical candela distribution, prismatic window, symbols 342.2T8 and 342.3T8.

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
0	910	910
5	904	910
15	872	906
25	810	881
35	713	817
45	577	709
55	378	566
65	195	381
75	88	227
85	30	119
90	9	76

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
95	8	48
105	6	26
115	5	22
125	3	15
135	1	9
145	0	4
155	0	1
165	0	0
175	0	0
180	0	0

TABLE III. Numerical candela distribution (red mode), one red filter center lamp energized, two white lamps off, symbols 333.1T8RC and 341.4T8.

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
0	1.73	1.73
5	1.73	1.68
15	1.63	1.44
25	1.44	0.78
35	1.2	0.22
45	0.92	0.18
55	0.62	0.13
65	0.33	0.08
75	0.12	0.05
85	0.03	0.03
90	0.02	0.02

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
95	0.01	0.01
105	0.01	0.01
115	0.01	0.01
125	0.01	0.01
135	0.01	0.01
145	0.01	0.01
155	0	0.01
165	0	0
175	0	0
180	0	0

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TABLE IV. Numerical candela distribution (white mode), two white lamps energized, red center lamp off, symbols 333.1T8RC and 341.4T8.

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
0	443	443
5	440	443
15	421	449
25	385	424
35	332	367
45	267	296
55	195	250
65	124	211
75	62	147
85	17	84
90	6	58
95	6	40
105	5	23
115	3	18
125	2	13
135	1	9
145	1	5
155	0	2
165	0	0
175	0	0
180	0	0

TABLE V. Numerical candela distribution, three white lamps energized, symbols 341.1T8 and 341.2T8.

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
0	608	608
5	540	551
15	338	350
25	116	156
35	20	27
45	8	8
55	4	3
65	1	1
75	0	0
85	0	0
90	0	0

PHOTOMETRIC DATA:

Photometry reports based on LM-41 and other pertinent Illuminating Engineering Society (IES) procedures.

Plane:	0 degrees is parallel to lamps, 90 degrees is perpendicular to lamps.
Electrical values:	120 volts, 60 hertz, 53.7 watts input power (three lamps energized).
Ambient temperature:	77±1.8 °F (25±1 °C).
Lamps:	GE F17T8-SP41-ECO, or equal (seasoned for at least 100 hours).

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Lamps correction:	Lamps rated at 1,325 lumens.
Ballast:	Furnished with fixture, see MIL-DTL-16377/81 for specification.

REQUIREMENTS:

CLASSIFICATION:

Type: I
Class: 2

SYMBOL NUMBERS:

333.1T8: Complete lighting fixture assembly, standard mounting with three 17-watt T8 cool white fluorescent lamps and white diffusing window assembly.

333.2T8: Complete lighting fixture assembly, flush mounting with three 17-watt T8 cool white fluorescent lamps and white diffusing window assembly.

341.1T8: Complete lighting fixture, standard mounting blue and white (Command Information Center [CIC] lighting) clear black shielded with three 17-watt T8 cool white fluorescent lamps.

341.2T8: Complete lighting fixture, flush mounting blue and white CIC lighting clear black shielded, with three 17-watt T8 cool white fluorescent lamps.

333.1T8RC: Complete lighting fixture assembly, standard mounting red filter on center lamp, center lamp shielded (berthing area lighting) with three 17-watt T8 cool white fluorescent lamps and white diffusing window assembly.

341.4T8: Complete lighting fixture assembly, flush mounting red filter on center lamp, center lamp shielded (berthing area lighting) with three 17-watt T8 cool white fluorescent lamps and white diffusing window assembly.

342.2T8: Complete lighting fixture assembly, standard mounting with three 17-watt T8 cool white fluorescent lamps and clear and white prismatic window assembly.

342.3T8: Complete lighting fixture assembly, flush mounting with three 17-watt T8 cool white fluorescent lamps and clear and white prismatic window assembly.

DESIGN:

Electrical rating:	51 watts, 120 volts, 60 hertz.
Dimension and configurations:	See figures 1, 2, 3, 6, 7, 8, 10, and 11.
Photometric data:	See figures 13, 14, 15, 16, and 17 and tables I, II, III, IV, and V.
Wiring diagram:	See figure 12 for typical wiring.
Ballast:	See MIL-DTL-16377/81 for specification.
Mounting dimensions:	
Symbols 333.1T8, 341.1T8, 333.1T8RC, and 342.2T8:	Standard mounting (surface), see figure 5.
Symbols 333.2T8, 341.2T8, 341.4T8, and 342.3T8:	Flush mounting (recessed), see figures 4 and 9.
Window assembly:	
Symbols 333.1T8 and 333.2T8:	White window, see table VI and figure 27 of MIL-DTL-16377.
Symbols 341.1T8 and 341.2T8:	White window with louvers, see table VI and figures 7, 8, and 9.
Symbols 333.1T8RC and 341.4T8:	White window with center lamp shielded, see table VI and figures 10 and 11.
Symbols 342.2T8 and 342.3T8:	Prismatic window, see table VI and figure 27 of MIL-DTL-16377.
Enclosure:	Watertight, see 3.5.11 of MIL-DTL-16377 dripproof (15 degrees) for 341.1T8 and 341.2T8.
Cable:	10 feet (minimum) of Type LSTHOF-3 in accordance with MIL-DTL-24643/3.

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Weight:

Symbols 333.1T8 and 342.2T8:	Not to exceed 15 pounds 11 ounces.
Symbols 333.2T8 and 342.3T8:	Not to exceed 17 pounds 14 ounces.
Symbol 341.1T8:	Not to exceed 19 pounds 4 ounces.
Symbol 341.2T8:	Not to exceed 21 pounds 7 ounces.
Symbol 333.1T8RC:	Not to exceed 16 pounds.
Symbol 341.4T8:	Not to exceed 18 pounds.

Designation and marking: The following labels shall be furnished in accordance with MIL-DTL-16377 and applied as shown on figures 1, 3, and 8:

a. Two Type I labels, equipment identification, reference MIL-DTL-16377 figure 11; shall include manufacturer and address, one located on back and other located on an inner nonreflecting surface.

b. One Type II label, circuit wiring information, reference MIL-DTL-16377 figure 12; located on outside surface adjacent to cable entrance.

c. One Type IV label, repair parts, reference MIL-DTL-16377 figure 14; information to be shown shall be "USE THE FOLLOWING REPAIR PARTS ONLY:" followed by a list of all standard parts shown in table VI, located on an inner nonreflecting surface.

Standard parts: The standard parts shown in table VI shall be used in the construction of the fixture specified herein.

EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and the following:

First article testing/inspection:

Complete fixtures:	See table VII.
Window assemblies:	See table VIII.

Conformance:

Comparison inspection:	See tables VII and VIII.
Conformance inspection:	See MIL-DTL-16377.

GENERAL INFORMATION:

Assignment of military part numbers and general information is shown in table IX.

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TABLE VI. Standard parts.

Name	Military P/N or Specification	National Stock Number (NSN)	Remarks
Lamp	Commercial identification F17T8/CW	6240-01-391-6027	
Lamplock	M16377/40-003		
Lampholder	UL 542	6250-00-284-0481	Commercial
Ballast	M16377/81-001		Integrated starter type
High-temperature ballast	M16377/81-002		High-temperature applications
High-temperature starter	FS-4	6250-00-179-2354	High-temperature applications
Startersocket	UL 496	6250-00-295-2738	High-temperature applications
Shock mount	M16377/41-008	6210-01-393-4302	0.067 steel, zinc plated
Window assembly, white diffusing	M16377/12-001	6210-00-295-2849	1/
Window assembly, clear black shielded	M16377/12-003	6210-00-725-9030	1/
Window assembly, clear and white prismatic	M16377/12-004	6210-00-814-5914	1/
Window assembly, with center lamp shielded			1/
Filter, 17-watt T8 fluorescent lamp (Dark Adaptation)	M16377/42-010	6210-01-313-8169	Red transparent
Filter, 17-watt T8 fluorescent lamp (1% transmission gray)	M16377/42-011		
Filter, 17-watt T8 fluorescent lamp (2.5% transmission gray)	M16377/42-012		
Filter, 17-watt T8 fluorescent lamp (Amber)	M16377/42-015		
Filter, 17-watt T8 fluorescent lamp (Broad band blue)	M16377/42-016		

NOTE:
1/ Window assemblies are the same as T12 lamp fixtures (MIL-DTL-16377/12).

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TABLE VII. First article tests and comparison inspection for complete fixtures. 1/

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Electromagnetic interference	3.6.14	4.8.16	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	2/
Voltage transient spike	3.6.18.2	4.8.23	3/
Dielectric withstanding voltage	3.6.15	4.8.2	4/
Insulation resistance	3.6.16	4.8.3	4/
Enclosure effectiveness	3.5.11	4.8.14.4	4/ Watertight, dripproof for 341.1T8 and 341.2T8.
Shock	3.5.12	4.8.7	4/
Vibration	3.5.13	4.8.8	4/
Enclosure effectiveness	3.5.11	4.8.14.4	4/ Watertight, dripproof for 341.1T8 and 341.2T8.
Dielectric withstanding voltage	3.6.15	4.8.2	4/
Insulation resistance	3.6.16	4.8.3	4/
Noise	3.5.14	4.8.9	
Continuity of grounding	3.6.13.4	4.8.12	
Light output	3.7.1	4.8.6.1	
Optical uniformity	3.7.6.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Magnetic permeability	3.4.8	4.8.18	

NOTES:
1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.
2/ If less, then 1 KVA exception not applicable for input wave form total harmonic distortion.
3/ Solid-state components shall be subject to 250 positive spikes and 250 negative spikes (500 total spikes), one spike per minute, using test method identified in MIL-STD-1399-300-1.
4/ Tests shall be performed in the order specified.

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TABLE VIII. First article tests and comparison inspection for window assemblies. [1/]

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.6.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Shock	3.5.12	4.8.7	[2/]
Vibration	3.5.13	4.8.8	[2/]
Enclosure effectiveness	3.5.11	4.8.14.4	[2/] Watertight, dripproof for 341.1T8 and 341.2T8.
Light output	3.7.1	4.8.6.1	Candela distribution

NOTES:
[1/] Requirement and test method paragraph numbers refer to MIL-DTL-16377.
[2/] Tests shall be performed in the order specified.

TABLE IX. Complete fixture part numbers.

Military Part Number M16377/80-	Item Name	National Stock Number (NSN)
333.1T8	Complete fixture, Symbol 333.1T8	6210-01-592-2397
333.2T8	Complete fixture, Symbol 333.2T8	
341.1T8	Complete fixture, Symbol 341.1T8	6210-01-592-2448
341.2T8	Complete fixture, Symbol 341.2T8	
341.4T8	Complete fixture, Symbol 341.4T8	
342.2T8	Complete fixture, Symbol 342.2T8	6210-01-592-2402
342.3T8	Complete fixture, Symbol 342.3T8	

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Custodians:
Army – CR4
Navy – SH
Air Force – 03

Preparing activity:
Navy − SH
(Project 6210-2022-004)

Review activities:
Navy – AS, CG
DLA – GS2

Civil agency:
GSA – FAS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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